



# RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

## Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

**Degree, Semester & Branch** : VII Semester B.E CSE  
**Course Code & Title** : AI3021 & IT in Agricultural System  
**Name of the Faculty member** : Mrs. S. Vijaya Amala Devi, AP / CSE

### Innovative Practice Description

- **Unit / Topic:** Unit I/Environmental Control Systems/Crop Growth
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 13
- **Activity Chosen:** Peer Learning
- **Justification:**
  - Crop growth involves several interconnected biological and environmental processes. Through peer teaching, these complex ideas become more accessible, practical, and memorable, as students explain concepts to one another using simple language and real-life examples. This peer-driven approach strengthens comprehension and promotes collaborative exploration.
  - Peer teaching encourages students to actively engage with the topic rather than passively listen. During discussions, learners visualize how environmental factors such as temperature, sunlight, soil nutrients, and water availability influence plant development, leading to deeper conceptual clarity.
  - Students gain confidence when explaining technical terms like germination, photosynthesis, and nutrient cycles to their peers. This process enhances their communication skills, reinforces scientific vocabulary, and nurtures a sense of responsibility in sharing knowledge.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**

The instructor began the session by providing a **brief overview of crop growth concepts, including germination, photosynthesis, nutrient cycles, and environmental factors affecting plant development.** This ensured that all students had a foundational understanding before the peer teaching session. Figure 1: Shows a student actively discussing crop growth concepts with peers, highlighting engagement and collaborative learning during the peer teaching session.

Based on students' interest and willingness, **Ms. Annalakshmi** was selected as the **peer teacher** for the session. Figure 2: Shows Ms. Annalakshmi presenting her group's ideas on crop growth, using visual aids and examples to explain key concepts to the class.

The peer teacher started by introducing different stages of crop growth and explaining how **biological processes and environmental conditions interact** to influence development:

- **Germination** – explaining the conditions required for seeds to sprout.
- **Photosynthesis** – showing how sunlight, water, and CO<sub>2</sub> contribute to plant energy production.
- **Nutrient cycles** – discussing the role of nitrogen, phosphorus, and potassium in crop health.
- **Environmental factors** – explaining how temperature, soil moisture, and light intensity affect growth.

The peer teacher conducted a **real-time demonstration using plant models and visual aids**:

- Highlighted germination stages using seed samples.
- Illustrated photosynthesis through diagrams showing energy conversion.
- Demonstrated nutrient absorption and the effects of deficiencies using sample crop images.
- Discussed the impact of environmental stress (like drought or excess water) on plant development.

Students were actively engaged in the session. The peer teacher **asked classmates to predict how changes in conditions** (like reduced water or nutrient deficiency) would affect crop growth, promoting discussion and critical thinking. Teams also discussed real-life applications such as irrigation scheduling, soil management, and crop rotation strategies.

The instructor concluded the session by **appreciating the peer teacher's effort**, providing feedback, and summarizing the key takeaways: the connection between theoretical crop growth principles and practical agricultural practices. The activity enhanced student **understanding, confidence, and collaborative learning**, making complex biological and environmental concepts more tangible and memorable.

**CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO5 | PO8 | PO10 | PO12 | PSO3 |
|-----|-----|-----|-----|-----|------|------|------|
| CO4 | 3   | 3   | 3   | 3   | 3    | 3    | 3    |

(1 – Low      2 – Moderate      3 – High)

• **PO / PSO mapped:**

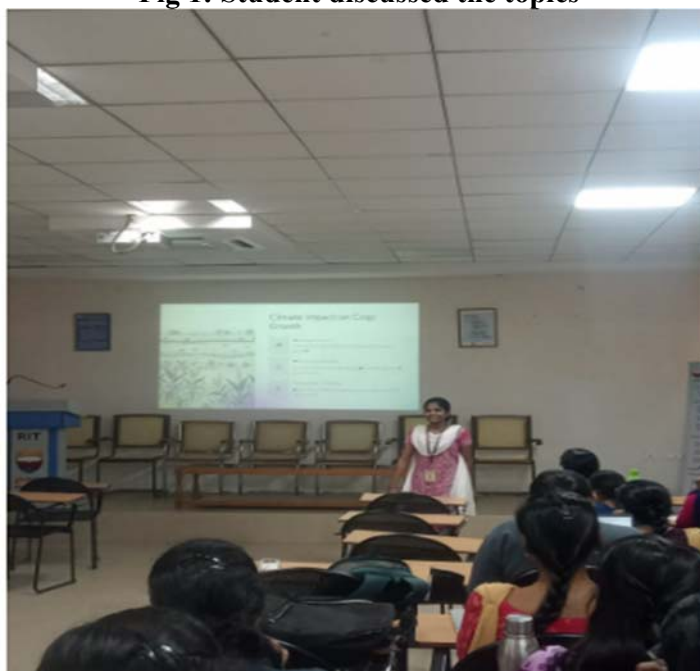
| Innovative practice                  | PO1                        | PO2                          | PO4                        | PO5                                              | PO10                                                | PO12                                              | PSO2                                                                 |
|--------------------------------------|----------------------------|------------------------------|----------------------------|--------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------|
|                                      | 3                          | 3                            | 3                          | 1                                                | 2                                                   | 2                                                 | 2                                                                    |
| <b>Justification for correlation</b> | Student apply the basic IT | Student identify and analyze | Student able to create the | Student evaluate the online measurement of plant | Student to present case studies on Plant production | Student to explore emerging applications for Crop | Students can use IoT devices and secure cloud platforms to implement |

|  |                                          |                                                                   |                                             |                                          |          |                        |                                                 |
|--|------------------------------------------|-------------------------------------------------------------------|---------------------------------------------|------------------------------------------|----------|------------------------|-------------------------------------------------|
|  | knowledge for Crop growth in Greenhouses | the problems in plant production and provide the solutions by IT. | basic concepts of GPS and mapping software. | growth in environmental control systems. | systems. | growth in greenhouses. | automated environmental control in greenhouses. |
|--|------------------------------------------|-------------------------------------------------------------------|---------------------------------------------|------------------------------------------|----------|------------------------|-------------------------------------------------|

• **Images / Screenshot of the practice:**



**Fig 1: Student discussed the topics**



**Fig 2: Annalakshmi presented their ideas**

**Reflective Critique:**

❖ **Feedback from Students and Stakeholders:**

- Students appreciated the opportunity to **explore crop growth concepts independently** before the peer teaching session.
- Access to pre-class materials allowed them to review stages like germination, photosynthesis, nutrient cycles, and environmental influences at their own pace.
- During the session, discussing these concepts with peers helped them **clarify doubts, reinforce understanding, and apply theoretical knowledge to practical scenarios** such as irrigation, nutrient management, and crop monitoring.

#### ❖ **Benefits of the Practice:**

- Peer teaching encouraged students to actively communicate and work together. Group discussions and team-based explanations fostered collaboration, helping learners clarify doubts, exchange ideas, and support one another's understanding of complex crop growth concepts.
- By teaching and discussing topics such as germination, photosynthesis, nutrient cycles, and environmental influences, students developed a deeper conceptual grasp. Explaining ideas to peers reinforced their own knowledge and made theoretical concepts more tangible and easier to remember.
- The use of diagrams, seed samples, images of nutrient deficiencies, and environmental stress scenarios made learning more interactive and practical. Students were able to relate theoretical knowledge to real-world agricultural practices, such as crop monitoring, irrigation planning, and sustainable farming techniques, which increased interest and motivation.

#### ❖ **Challenges Faced:**

- Not all students were willing to take on the role of peer teacher, which limited the number of active presentations during the session. This made it challenging to ensure that all students had equal opportunities to lead discussions and share their understanding.
- Some students struggled with understanding or correctly using technical terms related to crop growth, such as *germination*, *photosynthesis*, and *nutrient cycles*. These students required extra guidance from the peer teacher or faculty to fully grasp the concepts, highlighting the need for careful scaffolding and support during peer-led sessions.

#### **References:**

- ❖ [https://en.wikipedia.org/wiki/Peer\\_learning](https://en.wikipedia.org/wiki/Peer_learning)
- ❖ [Peer to Peer Learning - Examples, Benefits & Strategies](#)
- ❖ <https://whatfix.com/blog/peer-to-peer-learning/>

**Signature of Faculty Member**

**HOD**